

OLIST BRAZILIAN E-COMMERCE DATASET

Delivery Performance Audit

Last Mile Logistics Analysis

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Executive Summary

Business Question

The CEO of Veridi Logistics suspects that negative customer reviews are driven not just by late deliveries, but by **inaccurate estimated delivery dates** over-promising and under-delivering.

Key Findings

- **8.1%** of delivered orders arrive later than promised
- States AM, RR and PI have the highest late rates in the dataset
- On-time orders average **4.29/5** vs **1.78/5** for super-late orders
- Cross-state shipments underperform same-state by **0.3 pts** on review score

Bottom Line

Logistics *is* the problem. The data confirms the CEO's hypothesis: delivery accuracy directly drives customer sentiment. The issue is concentrated in states AL, MA and PI in cross-state shipments.

• Dataset

99,441 orders | **27** Brazilian states | **72** product categories | 2016–2018

Dataset & Methodology

Data Sources (Olist)

- `olist_orders_dataset.csv` as backbone
- `olist_order_reviews_dataset.csv` for sentiments
- `olist_customers_dataset.csv` for geography
- `olist_products_dataset.csv` for categories
- `olist_order_items_dataset.csv` for pricing
- `olist_sellers_dataset.csv` for s-origin
- `product_category_name_translation.csv`

Pipeline Summary

- ① Load & parse all CSVs (dates typed upfront)
- ② Deduplicate reviews to keep latest per order
- ③ Aggregate items to order level
- ④ Sequential LEFT JOINS (no row duplication)
- ⑤ Engineer delay, classification & time features
- ⑥ Null fill strategy: categorical → labelled; numeric delays → left as NaN
- ⑦ Export → Google Sheets → Looker Studio

• Tool: **Google Colab** | • Dashboard: **Google Looker Studio** | • Libraries: `pandas`, `numpy`, `matplotlib`

KPI Overview

99,441

Total Orders

All statuses

91.99%

On-Time Rate

Delivered orders

4.09

Avg Review Score

Out of 5.0

3.7%

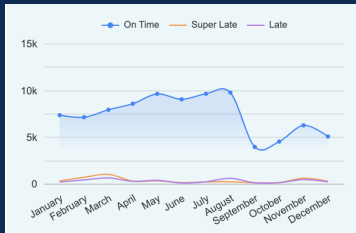
Late Orders

*1–5 days past
estimate*

4.4%

Super Late

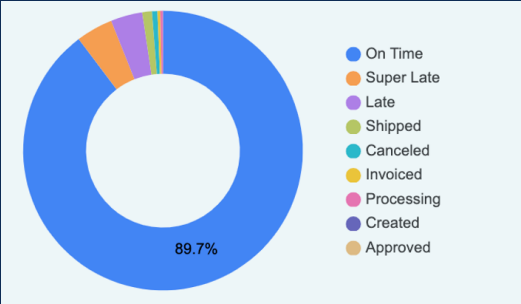
>5 days late



Reading the Numbers

- **On Time** orders dominate at **91.9%**
- **Late** (3.7%) + **Super Late** (4.4%) = **8.1%** failure rate
- Average delivery takes **11 days**
- Peak order months are from **May** to **August**

Delivery Status Breakdown



What This Shows

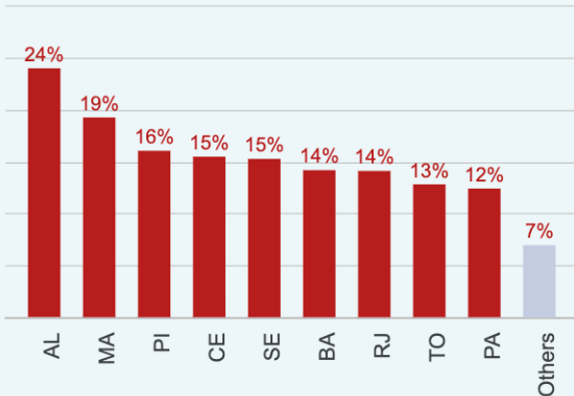
Orders split into three delivery outcomes:

Status	Count	%
On Time	88,644	89%
Late	3,615	3.7%
Super Late	4,211	4.3%

8.1% of all orders *miss* the promised delivery date, directly exposing customers to unmet expectations.

Geographic Analysis, Which States Are Failing?

Top 10 Late Rate | Cross States Shipments

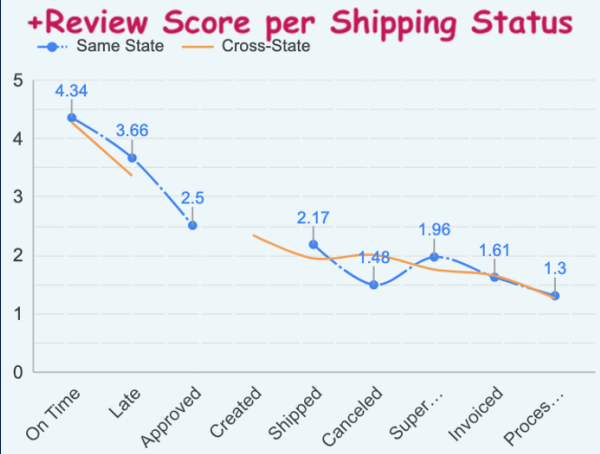


Top 5 States by Late Delivery Rate

Customer State	Late %
AL	24%
MA	19%
PI	16%
CE,SE	15%

Pattern: Some of the states with the highest late delivery rates are also among the states with the lowest order volumes suggesting these are underserved like SE.

Sentiment Correlation, Do Late Deliveries Cause Bad Reviews?



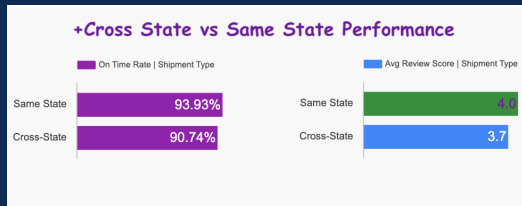
Review Score by Delivery Status

Status	Avg Review
● On Time	4.3 / 5
● Late	3.35 / 5
● Super Late	1.85 / 5

Gap = 2.45 points.

A customer who receives a late delivery is **significantly** more likely to leave a 1-star review. This directly answers the CEO: *“Fix logistics, fix reviews.”*

Cross-State Shipment Analysis



The Finding

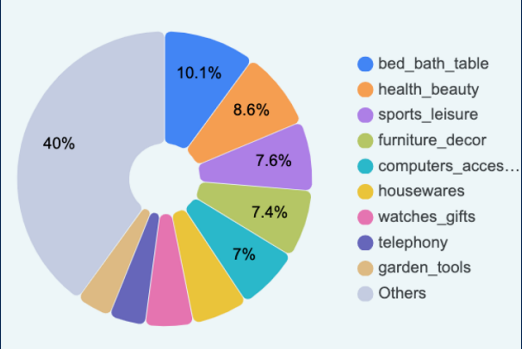
Type	On-Time %	Avg Review
Same State	93.93%	4
Cross-State	90.74%	3.7
Gap	3.19 pts	0.3 pts

Why This Matters

- **64%** of all orders are cross-state shipments
- Cross-state orders consistently underperform
- Root cause: long transit distances, more handoffs

Recommendation: Reducing the share of cross-state shipments through seller diversification or fulfilment expansion would directly improve both on-time rate and customer review scores, based on the gap shown in the data.

Product Category Insights



Top Categories by Volume

#	Category (EN)	Items
1	Bed, Bath & Table	11111
2	Health & Beauty	9483
3	Sports & Leisure	8396
4	Furniture & Decor	8171
5	Computers & Acc.	7679

All category names translated from Portuguese to English using `product_category_name_translation.csv`, making the dashboard fully readable for global stakeholders.

Recommendations & Conclusion

Three Actionable Recommendations

1. Prioritise States AL, MA and PI

These 3 states have the highest late delivery rates in the dataset. They should be the immediate focus for last-mile carrier review and service improvement.

2. Reduce Cross-State Shipment Dependency

64% of all orders are cross-state. Cross-state orders consistently show lower on-time rates and lower review scores than same-state orders.

3. Recalibrate Delivery Estimates

8.1% of orders miss the promised date. Adjusting estimated delivery dates to reflect actual delivery performance would immediately reduce the perceived failure rate.

Summary of Findings

- Late deliveries **directly** cause bad reviews, the data proves the CEO's hypothesis
- States AL, MA, PI have the worst late rates concentrated, not nationwide
- Cross-state shipments **consistently** underperform same-state on both metrics
- Category translation enables **global** stakeholder readability

[Dashboard Link](#)

[Notebook Link](#)